

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name the **three** components of the achalasia triad.
2. Which esophageal cancer arises from Barrett, and in **which third** of the esophagus?
3. What single feature separates a **Mallory-Weiss tear** from **Boerhaave syndrome**?

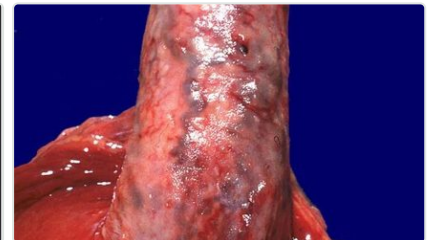
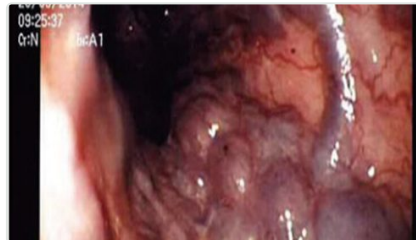
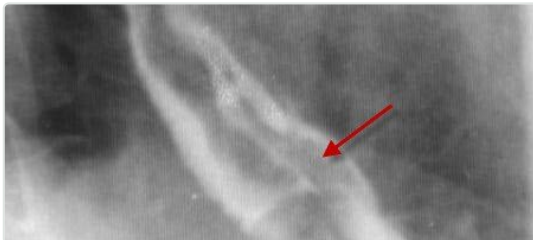
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Obstructive & vascular disease

Achalasia — definition	Motility disorder from absence of the myenteric plexus in the body of the esophagus → functional obstruction.
The triad	Increased LES tone + incomplete LES relaxation + esophageal aperistalsis. All three, every time.
Primary	Idiopathic — failure of distal esophageal inhibitory neurons.
Secondary	Chagas disease (<i>Trypanosoma cruzi</i>) destroys the myenteric plexus. The named cause to remember.
Achalasia-like	Diabetic autonomic neuropathy; infiltration by malignancy, amyloidosis, sarcoidosis; lesions of dorsal motor nuclei (polio, surgical ablation).

Esophageal varices

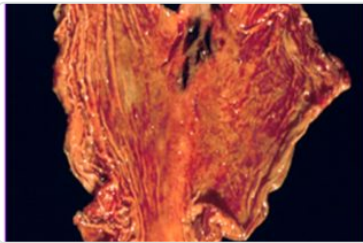
What	Dilated veins of the submucosal plexus of the distal esophagus , prone to rupture and haemorrhage.
Cause	Cirrhosis and portal hypertension in >90% , mostly alcohol-induced. Also hepatic schistosomiasis.
Mechanism	Gastro-oesophageal anastomoses link the lower oesophageal veins to the portal system. Portal hypertension makes them prominent, dilated, varicose.
The number	Varices >5 mm may rupture → life-threatening bleeding.
Mucosa	May be intact over an unruptured varix; ulcerated and necrotic once rupture has occurred.



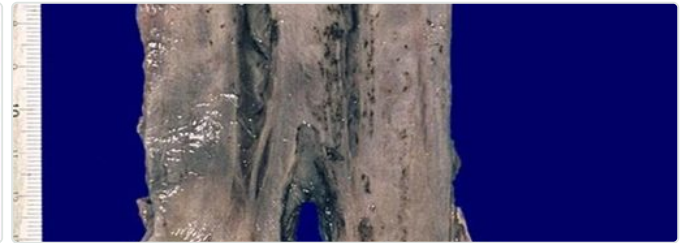
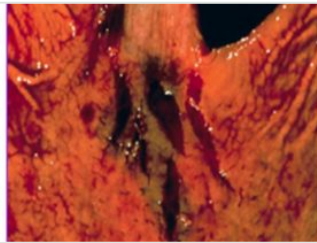
Barium swallow — serpiginous filling defects from varices Endoscopy — large varices at the GE junction Gross specimen — dilated submucosal veins

2 Lacerations — the depth is the discriminator

	Mallory-Weiss tear	Boerhaave syndrome
Depth	Mucosal tear	Transmural tear and rupture
Site	Longitudinal, near the GE junction	Distal esophagus
Trigger	Both: severe retching and vomiting , often acute alcohol intoxication — Boerhaave is the more severe end of the same mechanism. Failure of reflex relaxation of the GE musculature before the contractile wave.	
Presents with	GI bleeding, haematemesis	Severe chest pain, tachypnoea, shock
Consequence	Usually self-limiting	Severe mediastinitis — usually fatal



Mallory-Weiss — longitudinal mucosal tears at the GE junction



Transmural rupture — the Boerhaave end of the spectrum

3 Esophagitis — one finding separates each

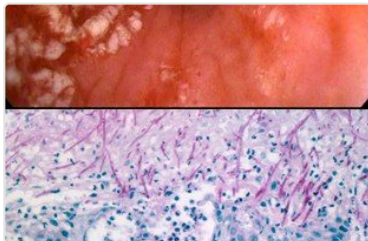
Full causes: reflux; chemical (alcohol, acids, alkali); medication (pill-induced, chemotherapy); infectious; eosinophilic; radiation; desquamative skin disease (bullous pemphigoid, epidermolysis bullosa); Crohn disease; graft-versus-host disease.

Type	Who	The identifying finding
Herpes (HSV)	Debilitated / immunosuppressed	Shallow punched-out vesicles and ulcers ; multinucleate squamous cells with nuclear inclusions
Candida	Debilitated / immunosuppressed	Adherent grey-white pseudomembranes of matted fungal hyphae
Eosinophilic	Atopic patients, peripheral eosinophilia	Transverse ridges resembling the trachea ; eosinophils far from the GEJ
Reflux (GERD)	Adults >40, also infants	Eosinophils then neutrophils ; basal zone hyperplasia , elongated lamina propria papillae

Eosinophils appear in both reflux and eosinophilic esophagitis — the discriminator is **number and location**: scattered and near the GEJ in reflux, numerous and far from the GEJ in eosinophilic esophagitis.



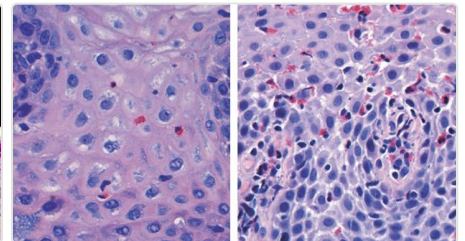
Herpes — punched-out ulcers



Candida — grey-white pseudomembranes



Eosinophilic — tracheal-like rings

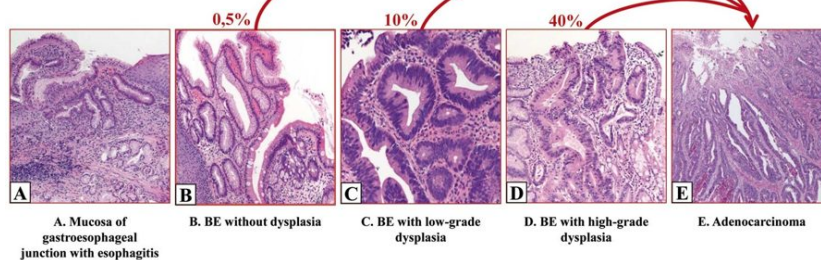


A reflux (scattered eos) · B eosinophilic (numerous)

4 GERD → Barrett → adenocarcinoma

This is the block's recurring pattern: chronic injury → metaplasia → dysplasia → carcinoma. You will meet it again as leukoplakia→SCC in the mouth, H. pylori→intestinal metaplasia→gastric cancer, adenoma→colorectal cancer, and PanIN→pancreatic cancer. Learn the shape once.

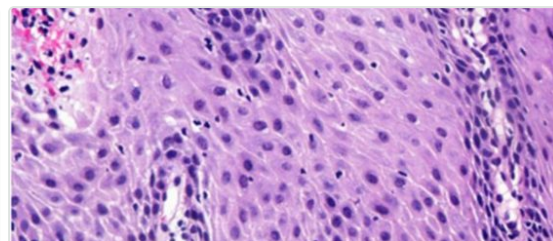
GERD causes	Anything that lowers LES tone or raises abdominal pressure : alcohol, tobacco, obesity, CNS antidepressants, pregnancy, hiatal hernia, delayed gastric emptying, increased gastric volume. Often no cause found.
GERD symptoms	Heartburn, dysphagia, regurgitation of sour gastric contents. Rarely severe chest pain mistaken for cardiac disease .
GERD complications	Ulceration · haematemesis and melaena · stricture · Barrett esophagus
Barrett esophagus	Acquired glandular metaplasia from long-standing GERD. Dysplasia is the intermediate stage between metaplasia and malignancy.



The progression: normal GEJ → Barrett without dysplasia → low grade → high grade → invasive adenocarcinoma



Barrett esophagus — salmon mucosa extending above the GEJ



Reflux histology — basal zone hyperplasia

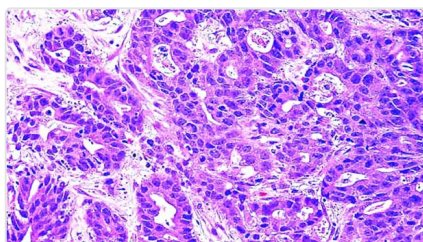
5 Adenocarcinoma vs SCC — the exam discriminator

Benign tumours first: epithelial (**papilloma**) and mesenchymal — **leiomyoma is the commonest benign tumour**. Almost all malignancies are epithelial.

	Adenocarcinoma	Squamous cell carcinoma
Where common	US and Western countries	More common worldwide
Site in esophagus	Distal third — may invade gastric cardia	Middle third in half of cases
Arises from	Barrett esophagus	Squamous epithelium directly
Trend	Incidence increasing worldwide	—
Risk factors	GERD → Barrett	Alcohol and tobacco · poverty · caustic injury · achalasia · Plummer-Vinson · very hot drinks · prior mediastinal radiation · nutritional deficiency · polycyclic hydrocarbons, nitrosamines, fungus-contaminated food · HPV in high-risk regions



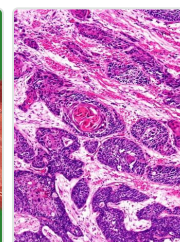
Adenocarcinoma — gross, distal esophagus



Adenocarcinoma — glandular histology



SCC — gross, middle third



SCC — histology

6 Clinical picture & spread

Presentation	Usually advanced at diagnosis . Dysphagia is the commonest symptom ; odynophagia; weight loss and debilitation; haemorrhage and sepsis; iron deficiency; aspiration via tracheo-oesophageal fistula .
Why it spreads	The rich submucosal lymphatic network promotes circumferential and longitudinal spread.
Tumour location	Lymph nodes involved
Upper third	Cervical LN
Middle third	Mediastinal, paratracheal, tracheobronchial LN

Tumour location	Lymph nodes involved
Lower third	Gastric and coeliac LN

Nodal spread follows the local lymphatics — the level of the tumour predicts the nodes. Easy marks if you can picture the anatomy.

P Must-remember pearls

- **Achalasia is a triad** — raised LES tone, incomplete relaxation, aperistalsis. Chagas is the named secondary cause.
- **Varices: cirrhosis and portal hypertension in over 90%**, and rupture risk climbs above 5 mm.
- **Mallory-Weiss is mucosal, Boerhaave is transmural** — same trigger, and the transmural one is usually fatal.
- **Barrett is the only acquired metaplasia here**, and dysplasia is the step between it and adenocarcinoma.
- **Adenocarcinoma = distal third, from Barrett, Western. SCC = middle third, worldwide.**
- **Dysphagia is the commonest symptom** of esophageal carcinoma — and it appears late.

Q Test yourself

1. A 46-year-old man with long-standing heartburn has an endoscopic biopsy showing glandular epithelium above the gastro-oesophageal junction. Which malignancy is he now at increased risk of, and where would it most likely arise?

- A. SCC of the middle third
B. Adenocarcinoma of the distal third
C. Leiomyoma of the lower third
D. SCC of the upper third

2. A patient from South America has dysphagia. Manometry shows raised LES tone, incomplete LES relaxation and aperistalsis. Which organism should be considered?

- A. *Candida albicans*
B. Herpes simplex virus
C. *Trypanosoma cruzi*
D. Human papillomavirus

3. Two patients vomit repeatedly after heavy drinking. One presents with haematemesis, the other with severe chest pain, tachypnoea and shock. What distinguishes the second patient's lesion?

- A. It is located in the proximal esophagus
B. It is transmural rather than mucosal
C. It is caused by a different mechanism
D. It is associated with portal hypertension

4. An immunosuppressed patient has odynophagia. Endoscopy shows adherent grey-white plaques over the mucosa. What is the most likely cause?

- A. *Candida albicans*
B. Herpes simplex virus
C. Eosinophilic esophagitis
D. Reflux esophagitis

5. A biopsy from the mid-esophagus of an atopic child shows numerous intraepithelial eosinophils far from the gastro-oesophageal junction, and endoscopy showed transverse ridges. Which feature most argues against reflux esophagitis?

- A. Presence of eosinophils at all
B. The patient's young age
C. Eosinophil location far from the GEJ
D. Basal zone hyperplasia

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • B — This is Barrett esophagus — acquired glandular metaplasia from chronic GERD, giving rise to adenocarcinoma, which characteristically arises in the distal third.

2 • C — Secondary achalasia in Chagas disease; *T. cruzi* destroys the myenteric plexus, producing the same triad as primary achalasia.

3 • B — Boerhaave syndrome is a transmural rupture causing mediastinitis; Mallory-Weiss is a mucosal tear. Same trigger, different depth, very different outcome.

4 • A — Candidal esophagitis — adherent grey-white pseudomembranes of matted hyphae. Herpes gives punched-out vesicles and ulcers instead.

5 • C — Eosinophils appear in both conditions; their *number and distance from the GEJ* is what separates eosinophilic esophagitis from reflux.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

